

Practice Final

Q1

1. Show these two logical expressions are equivalent:

$$B \vee C \vee (B \wedge C) \vee (A \wedge B \wedge C)$$

$$(\neg A \vee \neg B \vee \neg C) \wedge (B \vee C)$$

A1

Check the truth tables agree.

Out[*n*]=

a	b	c	$b \vee c \vee (b \wedge c) \vee (a \wedge b \wedge c)$
True	True	True	False
True	True	False	True
True	False	True	True
True	False	False	False
False	True	True	True
False	True	False	True
False	False	True	True
False	False	False	False

Out[*n*]=

a	b	c	$(\neg a \vee \neg b \vee \neg c) \wedge (b \vee c)$
True	True	True	False
True	True	False	True
True	False	True	True
True	False	False	False
False	True	True	True
False	True	False	True
False	False	True	True
False	False	False	False

Q2

2. Find a linear recurrence for this sequence: 1,1,2,3,2,2,7,9,1,4,29,24

A2

$$x_{n+3} = x_{n+2} - 2x_{n+1} + 3x_n$$

In[*n*]:= FindLinearRecurrence[{1, 1, 2, 3, 2, 2, 7, 9, 1, 4, 29, 24}]

Out[*n*]= {1, -2, 3}

Q3

3. Modular arithmetic: Show there is no solution to $2x = 11 \pmod{18}$

A3

List possible solutions and see there are none

```
Out[ ]:=
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
0 2 4 6 8 10 12 14 16 0 2 4 6 8 10 12 14 16
```

or argue that $2x \pmod{18} = 2x - 18k$ must be even

Q4

4. What is the power set of $\{a, b, c\}$?

A4

All possible subsets

$\{\{\}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$

Q5

5. Make a Venn diagram showing $\neg(A \wedge B) \wedge \neg(A \wedge C) \wedge (\neg A \wedge \neg B \wedge \neg C)$

A5

Q6

6. Evaluate $h(2,2)$ given $\forall n,m \geq 0$ (non-negative integers)

$h(0,m) = 1$

$h(n+1, 0) = h(n, 2)$

$h(n+1, m+1) = 2 h(n+1, m)$

A6

```
In[ ]:= Do[h[0, m] = 1, {m, 0, 2}];
Do[h[n + 1, 0] = h[n, 2], {n, 0, 1}];
Do[h[n + 1, m + 1] = 2 h[n + 1, m], {n, 0, 1}, {m, 0, 1}];
Grid[Table[Row[{"h"[n, m], "= ", h[n, m]}], {n, 0, 2}, {m, 0, 2}]]

h[0, 0] = 1 h[0, 1] = 1 h[0, 2] = 1
Out[ ]:= h[1, 0] = 1 h[1, 1] = 2 h[1, 2] = 4
h[2, 0] = 4 h[2, 1] = 8 h[2, 2] = 16

In[ ]:= Clear[h]
```

Q7

7. System of Equations. Write in matrix form. Solve.

a)

$$5x + 7y + 9z = 2$$

$$2x + 5y + 5z = 8$$

$$5x + 5y + 6z = 4$$

$$x + y = 7$$

A7a

```
In[ ]:= Row[{MatrixForm[{{5, 7, 9}, {2, 5, 5}, {5, 5, 6}, {1, 1, 0}}],
  ".", MatrixForm[{x, y, z}], " = ", MatrixForm[{2, 8, 4, 7}]]]
```

$$\text{Out[]} = \begin{pmatrix} 5 & 7 & 9 \\ 2 & 5 & 5 \\ 5 & 5 & 6 \\ 1 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \\ 4 \\ 7 \end{pmatrix}$$

```
In[ ]:= Reduce[{{5, 7, 9}, {2, 5, 5}, {5, 5, 6}, {1, 1, 0}}.{x, y, z} == {2, 8, 4, 7}]
```

```
Out[ ]:= False
```

No Solutions

Q7b

b)

$$6x + 5y + 6z = 2$$

$$8y + 8z = 1$$

$$7x + 2y + 8z = 5$$

A7b

```
In[ ]:= Row[{MatrixForm[{{6, 5, 6}, {0, 8, 8}, {7, 2, 8}}],
  ".", MatrixForm[{x, y, z}], " = ", MatrixForm[{2, 1, 5}]]]
```

$$\text{Out[]} = \begin{pmatrix} 6 & 5 & 6 \\ 0 & 8 & 8 \\ 7 & 2 & 8 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$$

```
In[ ]:= Reduce[{{6, 5, 6}, {0, 8, 8}, {7, 2, 8}}.{x, y, z} == {2, 1, 5}]
```

$$\text{Out[]} = z == \frac{151}{232} \ \&\& \ y == -\frac{61}{116} \ \&\& \ x == \frac{7}{58}$$

Unique solution

Q7c

c)

$$6x + 9y + 8z = 0$$

$$9x + 2y = 7$$

A7c

```
In[ ]:= Row[{MatrixForm[{{6, 9, 8}}, {9, 2, 0}]],
           ". ", MatrixForm[{x, y, z}], " = ", MatrixForm[{0, 7}]}
```

$$\text{Out[]:= } \begin{pmatrix} 6 & 9 & 8 \\ 9 & 2 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 7 \end{pmatrix}$$

```
In[ ]:= Reduce[{{6, 9, 8}}, {9, 2, 0}].{x, y, z} == {0, 7}]
```

$$\text{Out[]:= } y = -\frac{14}{23} - \frac{24z}{23} \&\& x = \frac{21}{23} + \frac{16z}{69}$$

Infinite solutions, in this case one parameter z is free.

Q8

8. For $A = \begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix}$ what is the eigenvalue for the eigenvector $v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

A8

```
In[ ]:= {{2, 3}}, {-1, -2}].{1, -1}
```

```
Out[ ]:= {-1, 1}
```

Since $A v = -v$ the eigenvalue is $\lambda = -1$

Q9

9. Show that $A A^t$ is a symmetric matrix.

A9

A matrix is symmetric iff $B = B^t$

so it suffices to show that $B = B^t$ for $B = A A^t$

Check:

$$B^t = (A A^t)^t = A^{tt} A^t = A A^t = B$$

using the identities $(X Y)^t = Y^t X$ and $X^{tt} = X$

Q10

10. In order to fit this bivariate data to the quantum model $y = a e^x + b e^{-x}$ using least squares on $A x = b$, calculate A and b

Out[]:=

x	y
-0.5	6.8
-0.25	4.6
0.	5.1
0.25	5.7
0.5	4.4

A10

The parameters are $x = \begin{pmatrix} a \\ b \end{pmatrix}$

so the rows of A are $(e^{x[i]} \ e^{-x[i]})$

and the rows of b are $(y[i])$

```
In[ ]:= Table[{Exp[x], Exp[-x]}, {x, {- .5, - .25, 0, .25, .5}}] // MatrixForm
```

Out[]//MatrixForm=

$$\begin{pmatrix} 0.606531 & 1.64872 \\ 0.778801 & 1.28403 \\ 1 & 1 \\ 1.28403 & 0.778801 \\ 1.64872 & 0.606531 \end{pmatrix}$$

```
In[ ]:= {6.8, 4.6, 5.1, 5.7, 4.4} // MatrixForm
```

Out[]//MatrixForm=

$$\begin{pmatrix} 6.8 \\ 4.6 \\ 5.1 \\ 5.7 \\ 4.4 \end{pmatrix}$$

Q11

11. Given this matrix A

Out[]//MatrixForm=

$$\begin{pmatrix} 0.669066 & 0.450062 & 0.600905 & 0.275154 \\ 0.873963 & 0.328853 & 0.167913 & 0.273417 \\ 0.960839 & 0.527525 & 0.717818 & 0.0528061 \end{pmatrix}$$

a) Make an educated guess as to the matrix rank of A

A11a

A random 3x4 matrix will have rank 3

Q11b

b) Suppose this is the singular value decomposition of A.

$$\text{Out}[*]= \begin{pmatrix} -0.54 & 0.29 & 0.79 \\ -0.49 & -0.87 & -0.012 \\ -0.68 & 0.39 & -0.61 \end{pmatrix} \cdot \begin{pmatrix} 1.9 & 0. & 0. & 0. \\ 0. & 0.39 & 0. & 0. \\ 0. & 0. & 0.2 & 0. \end{pmatrix} \cdot \begin{pmatrix} -0.76 & -0.4 & -0.47 & -0.17 \\ -0.48 & 0.13 & 0.79 & -0.35 \\ -0.37 & 0.13 & 0.15 & 0.91 \\ 0.22 & -0.9 & 0.35 & 0.16 \end{pmatrix}$$

Approximate A by $X \cdot Y$ where the rank of X and the rank of Y are both 1.

A11b

In[*]:= **Normal**[**-0.76 -0.4 -0.47 -0.17**]

Out[*]= { { **-0.763865, -0.404026, -0.474314, -0.168223** } }

$$\text{Out}[*]= \begin{pmatrix} -0.54 \\ -0.49 \\ -0.68 \end{pmatrix} \cdot (1.9) \cdot (-0.76 \quad -0.4 \quad -0.47 \quad -0.17)$$

In[*]:= **MatrixForm**[{ { **-0.54** }, { **-0.49** }, { **-0.68** } }. { { **1.9** } }. { { **-0.76, -0.4, -0.47, -0.17** } }]

Out[*]//MatrixForm=

$$\begin{pmatrix} 0.77976 & 0.4104 & 0.48222 & 0.17442 \\ 0.70756 & 0.3724 & 0.43757 & 0.15827 \\ 0.98192 & 0.5168 & 0.60724 & 0.21964 \end{pmatrix}$$

Q12

12. If A is 3 by 4 and B is 5 by 4, and C is the concatenation of A and B, then what are the dimensions of C?

A12

The concatenation of two arrays is the joining of the two arrays.

3x4 concatenated with a 5x4

is 8x4 since 3+5=8